

Protein expression

1-Upstream processing

Using cells to start produce protein

-After Upstream processing we have downstream processing

- fungi and bacteria ..., it's a expression host will be used to express proteins.

-Bacteria grow in bioreactors

-protein that will be used as feed for farm animals or for production of beer.

Different type of expression system (expression hosts):

Fungi: are eukaryotic and capable of posttranslational modifications allow proper folding of proteins.

-many species of fungi are used as hosts for engineered Proteins.

Plant

-used for protein expression, and can be genetically engineered to produce specific proteins that do not occur naturally.

-the advantage of use plant that these process encourages rapid growth and reproductive rates.

-Tobacco plant- great choice for biotech protein production(Once genetic material is integrated a million new plant protein factories can fill the fields)

-Disadvantages using plants: not all proteins can be expressed in plants, plant have cell wall which makes purification difficult and the process of glycosylation is slightly different from animal cells.

Mammalian cell culture system = challenging

Mammalian cells required high purity types of nutrients (very expensive) and require growth factors ,vitamin ,amino acid.

- Nutritional requirements are complex
- Mammalian cells grow slowly
- Easy to contaminate
- Best choice for proteins destined to be used in humans

-Mammalian cell best choice to produce protein that will used in human. Because the mammalian cell culture systems are very similar to human cells.

Animal bioreactors production systems

We use it to produce protein.

We use mice and rabbits for example to produce monoclonal antibodies.

Mice are injected with an antigen and mouse secretes the desired antibody and the antibody is then purified

- What is an antibody? Proteins produced in reaction to antigens- invading viruses or bacteria

-Antibodies combine with and neutralize an antigen protecting the organism.

-Production of antibodies is part of immune response that helps living things resist infectious disease.

Insect system

Consists of insect cell's + viruses

-Baculoviruses (viruses that infect insects cell's) are used as vehicles to insert mammalian DNA causing the desired proteins to be produced by insect cells.

We can express any protein inside insects cell by cloning the gene including the protein in to a virus then we allow the recombinant virus infects the insect cells and your gene will be transmitted in to the insects cell by the virus, the insect cell will express your gene to produce your protein.

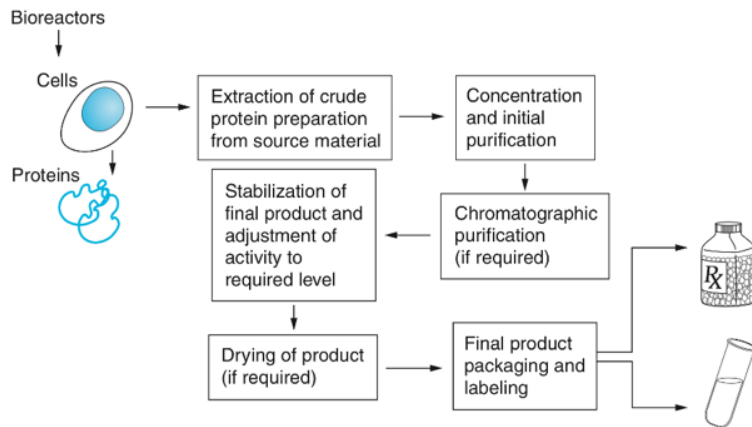
The problem that :

-Sometimes the posttranslational modification of proteins is slightly different in insects than in mammals.

The insect cell came from "silkworms" that used to produce protein.

-Currently used when small quantities of proteins are needed in research.

-insects cells have been used to produce protein that will used as vaccines for animals.



-Downstream processing methods

Contain

1-protein purification

2-Protein Verification

3-Protein formulation(the final stage)

Purification (depends on the protein)

We have a solution that contain protein only

Sometime we need higher pure protein (1% pure)

protein preparation sometime not, that depend In finally application\ uses.

Why the drugs(protein drugs)are expensive? Because they will require extensive purification. (More purification of your protein , more expensive products)

When you want to purify protein you need to use several techniques and maybe need to use very expensive substances. When using several purification steps, After each you will lose the activity of protein, so

some companies design purification steps very carefully to make sure we doing best purification will can and don't lose too much of protein and don't lose the activity of protein.

-During protein production you will need **chemical engineered** and **microbiologist** that under stand how the bacteria grows in medium , we need a **chemist** to understand the chemical composition of media and how the environment inside the fermenter changes, and the fermenter connected to software so we need a **computer software**. After production we need chemical engineer and biochemist and pharmacist.

So we have different people doing different stages.

Note:

Concentration :relating the amount of the solute to the volume of the solvent. Reduce the volume of the solvent in which my protein exists.

Protein purification

- Step 1: preparing an extract for purification

- Protein must be harvested(Protein harvesting\recovery)

Recovery means you want to separate your protein from the cellular waste we do that by Centrifugation\filtration.

-Entire cell is harvested if protein is intracellular requires cell lysis to disrupt the cell wall and release the protein. When you break open the cell we found something called crude extract or cell lysate. Then we can do centrifugation to separate the soluble proteins from the insoluble cell consistent or do filtration.

Cell lysate = cell opening

-Methods include freeze/thaw; detergents; mechanical methods mixture which releases the entire contents of the cell

*After cells are ruptured organic alcohols and salts are added to mixture

-agents increase the interactions between protein molecules to separate them from the mixture

-Culture medium is collected if the protein is extracellular because it is excreted into the culture medium.

summary

So here have

Intracellular protein production and extracellular protein production. If your protein inside the cell you have open the cell, if your protein not inside the cell(in medium)you don't need open it. collect the medium and remove the cell by Centrifugation or filtration, but you make sure that the cell is not open.

Different types of cell lead to different types of breaking cells.

To lysis the cell we use chemical techniques and physical techniques. We can use blender.

We have different methods to break the cell depending on : type of cell and type of protein will purifying and the final application of the protein.

We can use the detergent in producing a protein not enzyme because detergent make the enzyme inactivation.

Now the protein is outside the cel it not protected its need to protection. The protein now more likely to be damaged.

- Step 2: stabilizing proteins in solution

Protease:enzymes can degrade the protein.

Factors that might damage the protein: protease ,PH ,temperature ,oxidation, forming, bacterial contamination.

We take the protein and put them inside the buffer to detect them,this mixture contains: inhibition protein , buffering system (ph) ,antioxidant , antifoaming agents. Antibiotics.

Do blending in low temperature

- Maintain low temperature and proper pH in buffering solution

- Add protease inhibitors and antimicrobials to prevent proteins from being Digested that's Important to remove these additives later during purification process
- Add additives to prevent foaming and shearing from destroying the protein, These additives must be removed later.

Step 3 : separating the Components in the extract

Similarity between proteins permit us to separate them from material such us lipids, carbohydrates, and nucleic acids.

Proteins often have Hydrophilic amino acids on their surfaces that attract and interact with water molecules.

NOTE : we can use that characteristic as base for separating proteins from substance in the extract.

Salts, most common Ammonium sulfate, add to protein mixture to precipitate the proteins, solvents like ethanol, isopropanol, Acetone,... Also cause precipitate by remove water from between the proteins

Methods used protein separation.

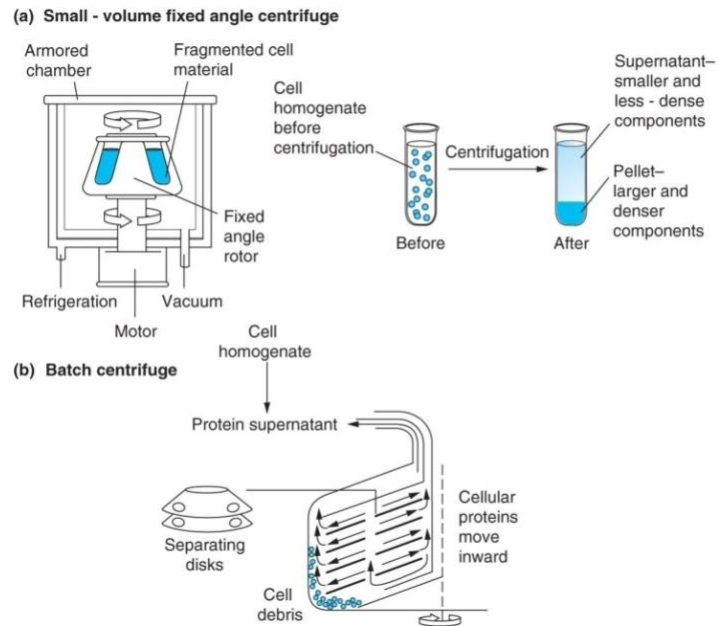
Filtration separation method (base on size)

1-Centrifugation : separate samples by spinning them at high speed.

With this process, proteins can often be isolated in a single layer or separated from heavier cell Components.

Two types of centrifugation :

- 1- Small volume centrifugation : processing only few liters at each run
- 2- Large reactor : process hundreds or thousands of liters.



2 - membrane Filters :

Filters of various Sizes and types can be used to Separating proteins from other molecules in the mixture.

Membrane filter : thin membrane of nylon or other engineered substances with varying pore Sizes are use to filter out all of the cellular debris from solution.

Two types of Filtrations

- 1- **Microfiltration** : removes the precipitates and Bacteria.

2- Ultrafiltration : can catch molecules such as proteins and nucleic Acids.

Note : some Ultrafiltration process can even separate large proteins from smaller one.

Problem of using membrane Filtration : Clog of filter membrane.

Diafiltration and Dialysis

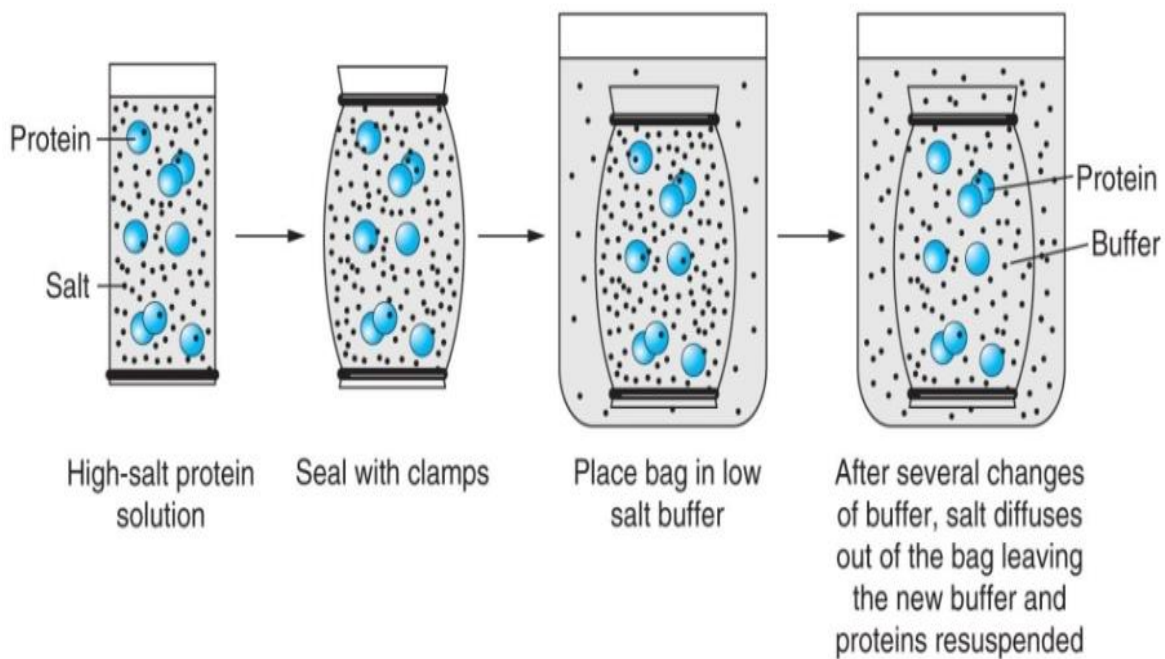
Are Filtration methods that rely on the chemical concept of equilibrium.

Dialysis depend on the ability of some molecules to pass through semipermeable membrane, other can not , because of their size.

Dialysis often require to remove the smaller Salts, solvents, and other additives used earlier in the purification.

Salts are then replaced with buffering agents that help the proteins during the remainder of the process.

Diafiltration : adds a filtering Component to Dialysis.



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